

Ruler vs Ruled: Monte Carlo Simulation Report

Ruler vs Ruled model created by the Pragmatic Realist (find me on Substack).

1. Introduction & Game Overview

This report presents the findings from an advanced Monte Carlo simulation of the 'Ruler vs. Ruled' dynamic. The model is structured as an escalating, multi-turn game of Chicken, where two actors (the Ruler and the Ruled) interact over a maximum of 1,000 steps per simulation.

In this environment, the 'Perceived Value of Society' (V_{res}) acts as a dynamic, shared resource representing general prosperity. During each step, both actors independently choose to act 'Hawkish' (aggressive/escalatory) or 'Dovish' (accommodating/de-escalatory). Hawkish actions extract a higher cost from the societal pool. Early termination of the simulation is triggered when systemic costs exhaust the total available value ($PERCEIVED_VALUE = 10.0$), indicating a structural societal failure (Revolution or Crackdown based on the distribution of pain).

2. Simulation Setup & Frameworks

The simulation sweeps across 8 distinct architectural frameworks to isolate the effects of differing strategic baselines and environmental sensitivities:

- Simulation 1 (Stochastic Baseline):** Completely random probability allocations at every step to map pure systemic noise.
- Simulation 2 (Blind Control):** Fixed ideological quadrants (DD, DH, HD, HH, RR) established at the start. These agents are perfectly oblivious to the decay of the societal resource pool.
- Simulation 3 (Neutral Players):** Random base hawkishness $[0, 1]$ penalized by conscious awareness of structural decay (Aversion Penalties).
- Simulation 4 (Neutral TFT):** Players reactively copy the other's previous move. (If opponent played Dove, P_{Hawk} drops to $[0, 0.5]$). If opponent played Hawk, P_{Hawk} jumps to $(0.5, 1]$).
- Simulation 5 (Aggressive Ruler):** Ruler has a fixed base hawkishness >0.5 ; Ruled remains neutral $[0, 1]$.
- Simulation 6 (Aggressive Ruler TFT):** Ruler remains strictly Aggressive (>0.5); Ruled drops the neutral posture and adopts Tit-for-Tat copying.

7. Simulation 7 (Aggressive Ruled): Ruler is neutral; Ruled has a fixed base hawkishness >0.5 .

8. Simulation 8 (Aggressive Ruled TFT): Ruler adopts TFT copying; Ruled remains strictly Aggressive (>0.5).

9. Simulation 9 (Blind TFT Control): A repeat of Simulation 2, but with both players using a Tit-for-Tat strategy, without responding to societal decay.

Strategic National Archetypes (Conscious Decay)

For simulations incorporating environmental decay awareness (Sims 3-8), we define five explicit aversion reactions:

1. Equals: Both Ruler and Ruled de-escalate collaboratively as resource pools drop (Linear 1:1 scaling).

2. Cowards: Symmetrically risk-averse, capitulating immediately into extreme dovish behavior under pressure (Quadratic scaling).

3. Fools: Symmetrically aggressive, resisting compromise and triggering stalemates (Sub-linear square-root scaling).

4. Tyrants: Hawkish Ruler (cushioned decay) and dovish Ruled (linear decay).

5. Brinksmen: Hawkish Ruled (cushioned decay) and dovish Ruler (linear decay).

3. State Failure Probability Analysis

This section details the exact probabilities of early state failure (and the corresponding triggers) across the strategic national groups. Each archetype is presented in its own table, with the blind Sim 2 Control and Sim 9 Control TFT included in every table as the comparative baselines.

Archetype: Equals

Configuration	Total Failure	Revolution (Ruler Lost)	Crackdown (Ruled Lost)
Sim 2 Control	63.41%	31.70%	31.70%
Sim 9 Control TFT	91.57%	45.79%	45.78%
Sim 3 Neutral	0.58%	0.29%	0.29%
Sim 4 Neutral TFT	0.08%	0.04%	0.04%
Sim 5 Agg Ruler	1.15%	0.58%	0.57%
Sim 6 Agg Ruler TFT	0.64%	0.32%	0.32%
Sim 7 Agg Ruled	1.14%	0.57%	0.57%
Sim 8 Agg Ruled TFT	0.64%	0.31%	0.32%

Archetype: Cowards

Configuration	Total Failure	Revolution (Ruler Lost)	Crackdown (Ruled Lost)
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		Lost)	Lost)
Sim 2 Control	63.41%	31.70%	31.70%
Sim 9 Control TFT	91.57%	45.79%	45.78%
Sim 3 Neutral	0.00%	0.00%	0.00%
Sim 4 Neutral TFT	0.00%	0.00%	0.00%
Sim 5 Agg Ruler	0.00%	0.00%	0.00%
Sim 6 Agg Ruler TFT	0.00%	0.00%	0.00%
Sim 7 Agg Ruled	0.00%	0.00%	0.00%
Sim 8 Agg Ruled TFT	0.00%	0.00%	0.00%

Archetype: Fools

Configuration	Total Failure	Revolution (Ruler Lost)	Crackdown (Ruled Lost)
Sim 2 Control	63.41%	31.70%	31.70%
Sim 9 Control TFT	91.57%	45.79%	45.78%
Sim 3 Neutral	23.00%	11.50%	11.50%
Sim 4 Neutral TFT	10.63%	5.30%	5.32%
Sim 5 Agg Ruler	41.87%	21.16%	20.71%
Sim 6 Agg Ruler TFT	47.51%	24.45%	23.06%
Sim 7 Agg Ruled	41.90%	20.73%	21.18%
Sim 8 Agg Ruled TFT	47.49%	23.06%	24.43%

Archetype: Brinksmen

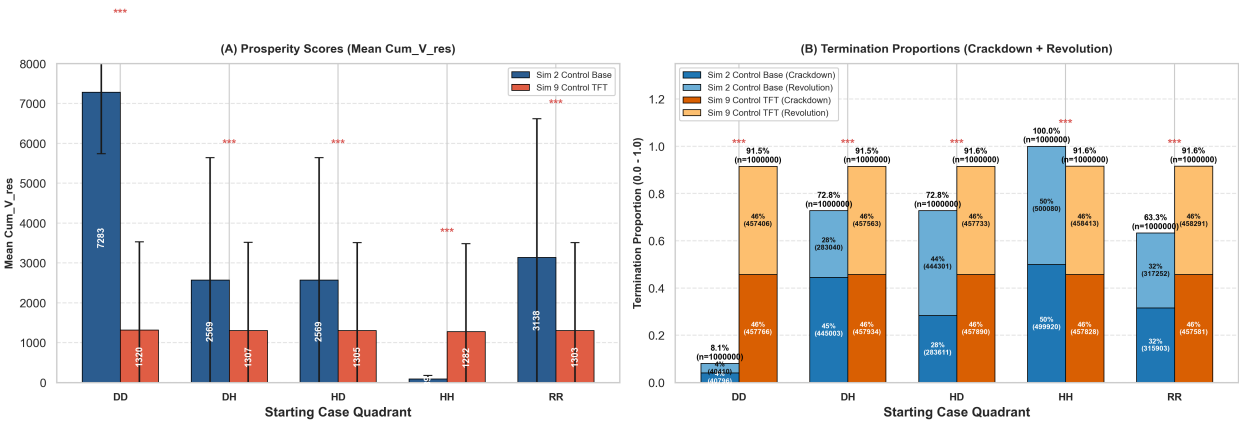
Configuration	Total Failure	Revolution (Ruler Lost)	Crackdown (Ruled Lost)
Sim 2 Control	63.41%	31.70%	31.70%
Sim 9 Control TFT	91.57%	45.79%	45.78%
Sim 3 Neutral	22.21%	5.60%	16.60%
Sim 4 Neutral TFT	6.29%	1.89%	4.40%
Sim 5 Agg Ruler	36.69%	9.78%	26.92%
Sim 6 Agg Ruler TFT	22.55%	7.07%	15.48%
Sim 7 Agg Ruled	43.09%	10.76%	32.33%
Sim 8 Agg Ruled TFT	56.05%	14.15%	41.89%

Archetype: Tyrants

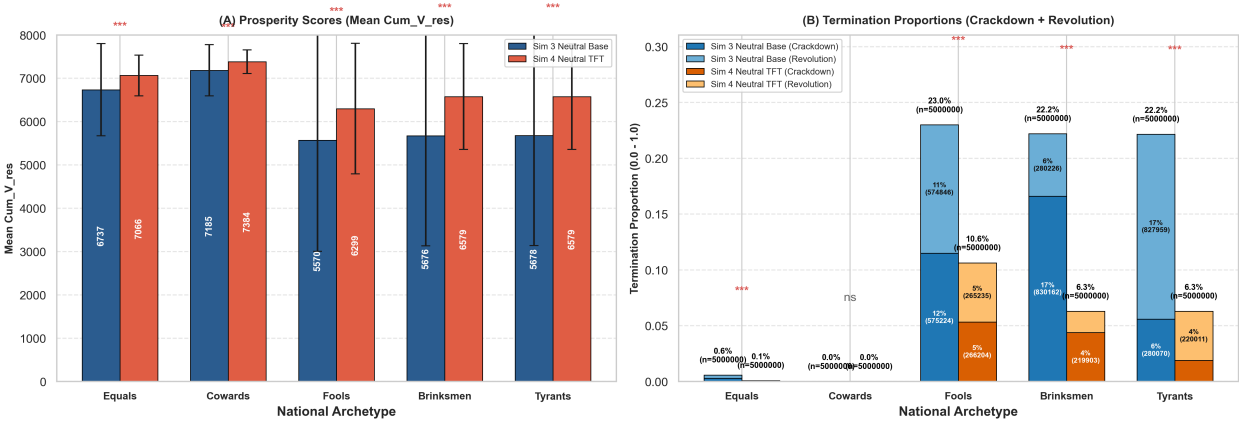
Configuration	Total Failure	Revolution (Ruler Lost)	Crackdown (Ruled Lost)
Sim 2 Control	63.41%	31.70%	31.70%
Sim 9 Control TFT	91.57%	45.79%	45.78%
Sim 3 Neutral	22.16%	16.56%	5.60%
Sim 4 Neutral TFT	6.29%	4.40%	1.89%
Sim 5 Agg Ruler	43.11%	32.34%	10.77%
Sim 6 Agg Ruler TFT	56.01%	41.86%	14.15%
Sim 7 Agg Ruled	36.67%	26.89%	9.78%
Sim 8 Agg Ruled TFT	22.54%	15.46%	7.08%

4. Analytical Visualizations

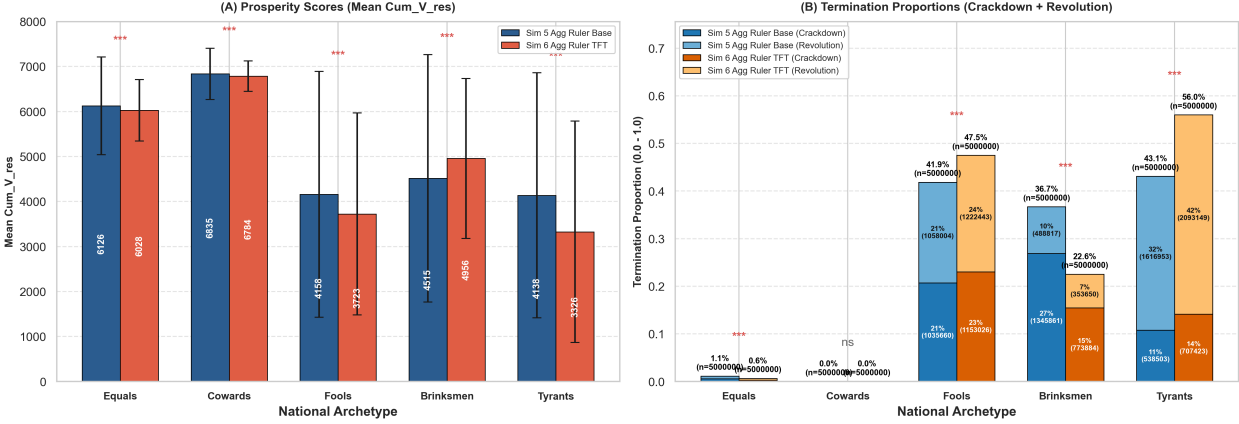
Paired Analysis: Control (Sim 2 vs 9)



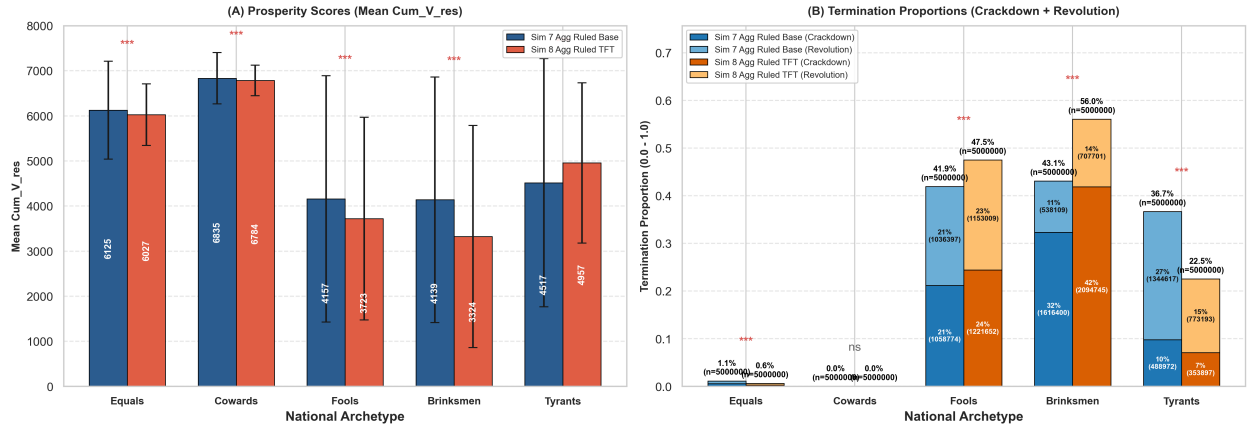
Paired Analysis: Neutral (Sim 3 vs 4)



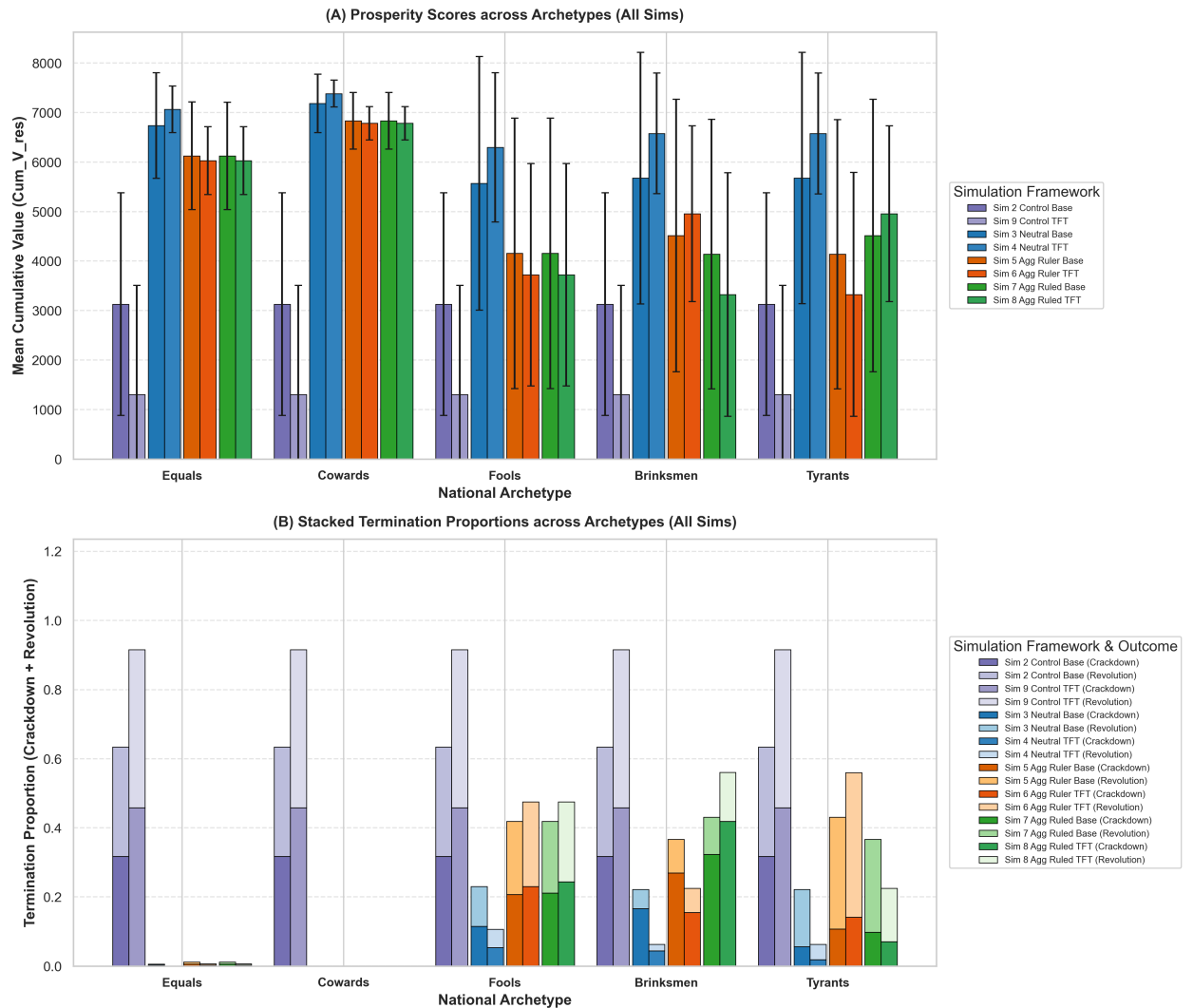
Paired Analysis: Aggressive Ruler (Sim 5 vs 6)



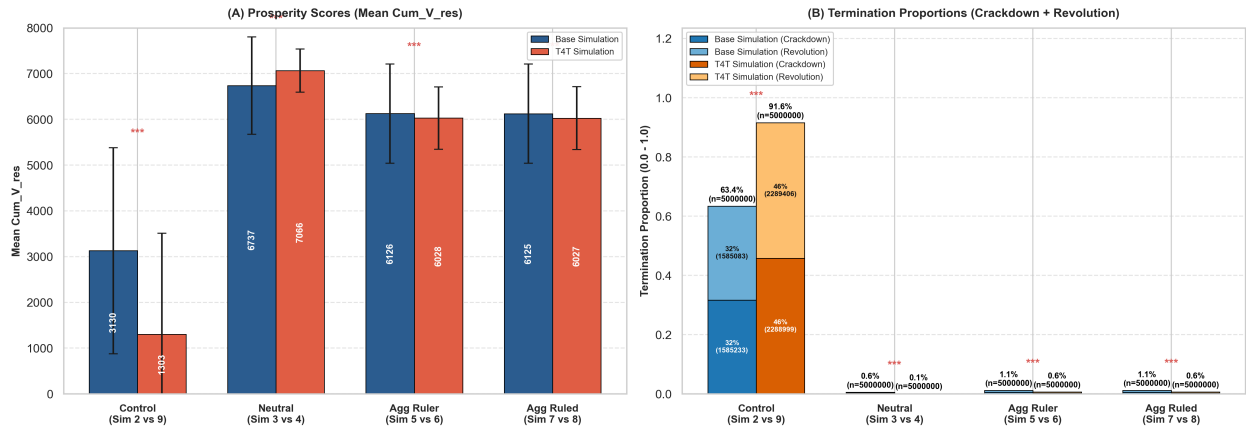
Paired Analysis: Aggressive Ruled (Sim 7 vs 8)



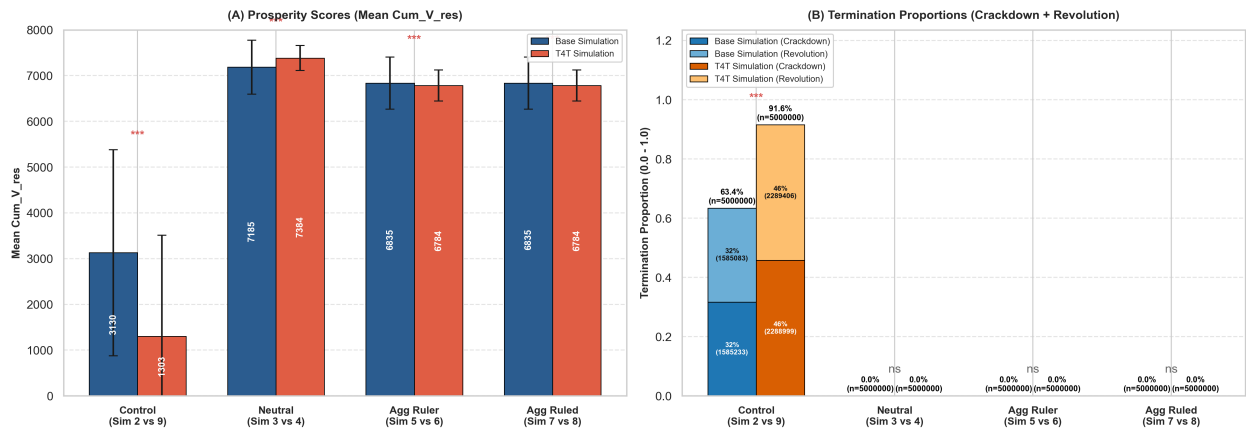
Consolidated Overview: Prosperity vs. Societal Stability across Archetypes



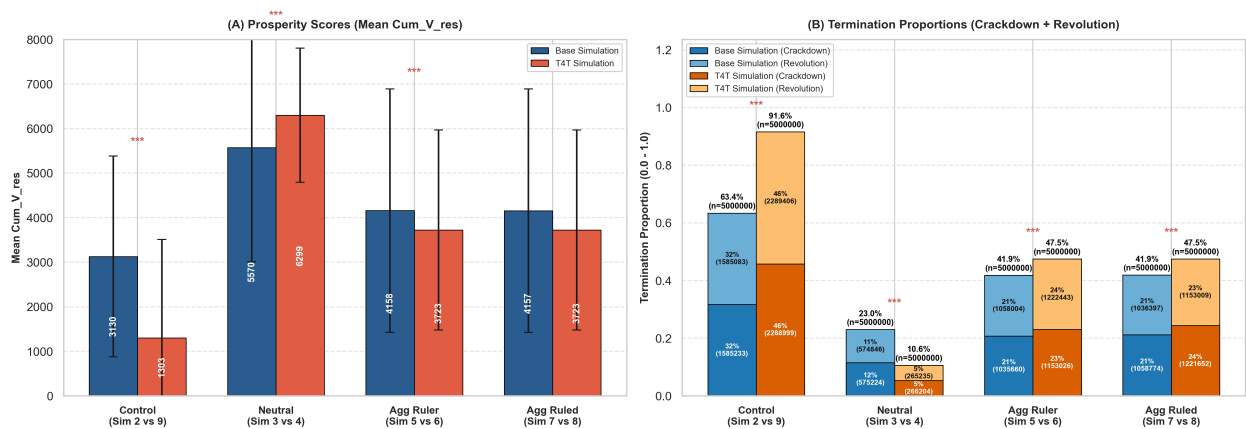
Archetype Paired Analysis: Equals



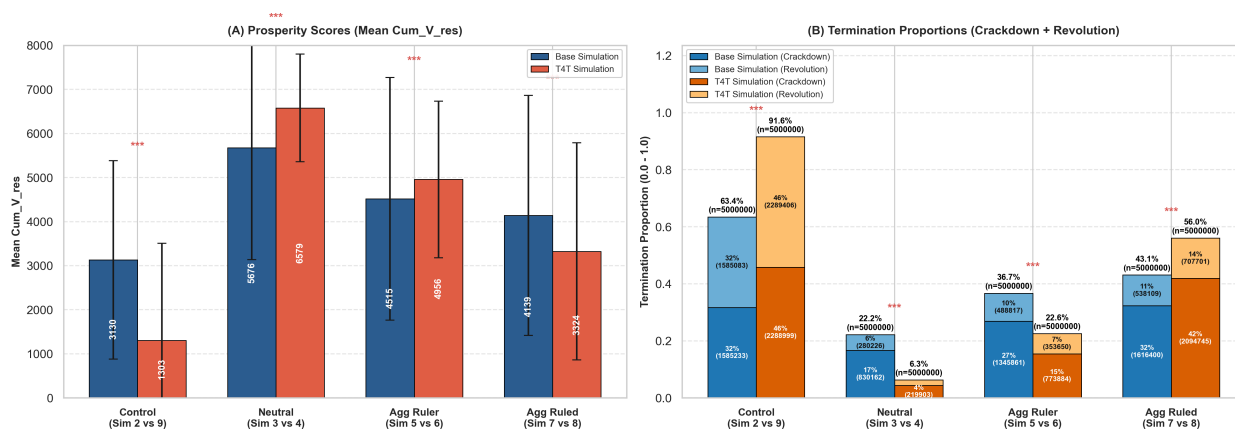
Archetype Paired Analysis: Cowards



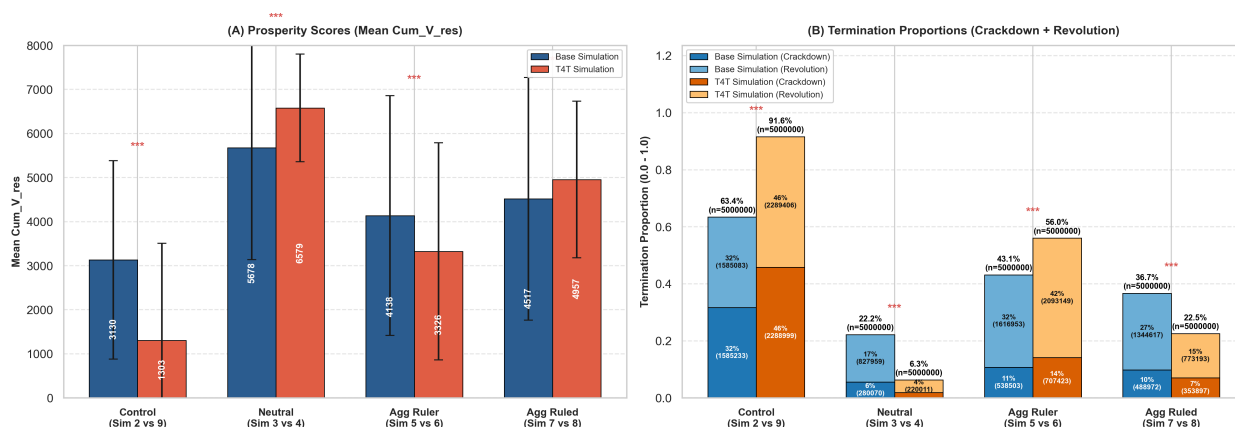
Archetype Paired Analysis: Fools



Archetype Paired Analysis: Brinksmen



Archetype Paired Analysis: Tyrants



5. Conclusion & Core Insights

Based on the comprehensive multi-framework analysis generated by this simulation run, several profound socio-structural dynamics emerge:

1. The Stabilizing Power of Tit-for-Tat (TFT): Across nearly all aggressive and unyielding archetypes (most notably the Fools and Tyrants), the introduction of a reactive, copy-based strategy dramatically curtails systemic collapse. By structurally guaranteeing retaliation against aggressive overreach, TFT functions as a powerful institutional governor, breaking runaway escalatory cycles before they drain the societal resource pool.

2. The Lethality of Strategic Blindness: The Sim 2 Control baseline operates with fixed ideological parameters and zero internal awareness of the wasting systemic value (V_res). This configuration consistently yields the highest rates of catastrophic collapse, which results in escalation spirals when Tit-for-Tat is introduced. This suggests that structural ignorance or political blindness to environmental/systemic decay is inherently more fatal to a society than

explicit aggression. If neither side are willing to, or capable of reacting to the decline of society, because of foolishness or propaganda, the probability of systemic collapse is maximised.

3. The Preservation Efficiency of Symmetrical De-escalation: The 'Cowards' (highly risk-averse) and 'Equals' (cooperative) archetypes consistently demonstrate near-perfect survival rates and maximize the prosperity metric. In a multi-round environment characterized by mutual destruction (the Hawk-Dove dynamic), rapid, symmetrical capitulation in response to resource scarcity effectively insulates the collective wealth from structural friction.

The Ruler vs Ruled Monte Carlo Simulation was created by the Pragmatic Realist: <https://thepragmaticrealist.substack.com>.